

Project Design Phase
Problem – Solution Fit Template

Date	03 JULY 2026
Team ID	
Project Name	Credit Card Approval Prediction System
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The Problem-Solution Fit simply means that you have found a problem with your customer and that the solution you have realized for it actually solves the customer’s problem. It helps entrepreneurs, marketers and corporate innovators identify behavioral patterns and recognize what would work and why

Problem

- Applicants face long waiting times and unclear approval criteria when applying for credit cards.
- Manual checks lead to inconsistent decisions and frustration among customers.
- Banks struggle with false approvals/rejections, increasing risk and lowering customer satisfaction.

Solution

- Build an AI-powered prediction system that automates approval decisions.
- Use machine learning models (Logistic Regression, Random Forest, XGBoost) trained on applicant data.
- Provide a web interface for applicants to submit and track applications.
- Ensure fairness and transparency by monitoring bias and accuracy.
- Deliver real-time status updates to reduce anxiety and improve trust.

Table:

Aspect	Problem	Solution
Customer Pain	Slow, unclear, manual approval process	Automated ML-based approval
Behavior	Applicants repeatedly call or reapply	Applicants track status online
Impact	Frustration, anxiety, dissatisfaction	Faster, fairer, transparent decisions
Business Challenge	High risk, poor efficiency	Improved efficiency, reduced risk
Adoption Trigger	Need for speed and fairness	AI system delivers instant predictions

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